



THE TEAM



GABRIEL CAVALCANTE

As team leader, I oversee the robot's structural design and overall development. I design the chassis, rescue mechanisms, and other components using CAD and 3D printing, focusing on compact and efficient solutions. I also support the assembly and integration of the robot's mechanical and electronic systems.



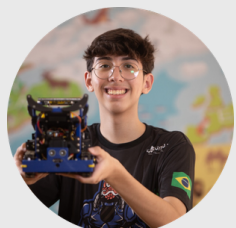
CAIO FIDELIS

As one of the lead programmers, I specialize in line-following algorithms and movement control systems. I improved navigation accuracy through PID-based tunings and adaptive movement strategies. I also refine the robot's performance on curves, intersections, ramps, and uneven surfaces to ensure reliable motion.



ARTHUR NOGUEIRA

Since joining Brainbots, I have worked on the robot's electrical integration and mechanical assembly. I am responsible for wiring, power management, and component integration. I also perform maintenance and repairs to ensure reliable performance during competitions.



PEDRO GOMES

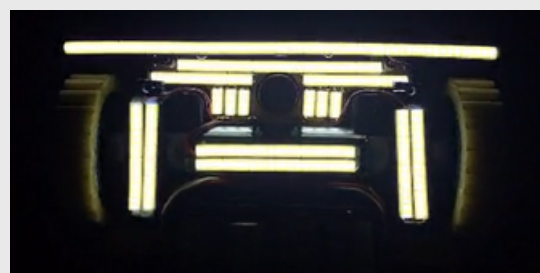
My primary responsibility is the robot's rescue room and computer vision system. I developed the rescue logic and vision architecture for victim recognition and autonomous decision-making. Using AI, OpenCV, and real-time object detection, I improved the robot's reliability during navigation and rescue tasks.

OUR ROBOT

CHARLES JR.

The robot features a custom 3D-printed PLA structure that integrates all mechanical and electronic components while reducing weight, simplifying maintenance, and allowing rapid design modifications. Its drivetrain uses 100:1 Metal Gearmotor 20Dx44L mm 6V CB motors and polyurethane rubber wheels (Shore 20, 30 mm radius x 20 mm width), providing reliable traction, stability, and precise maneuverability throughout the Rescue Line field. The modular design also facilitates the replacement and adjustment of components during testing and competition.

The robot uses a wide-angle camera (200°) dedicated to line following and an other (125°) to victim detection. Both are connected to a Raspberry Pi 5, which executes the computer vision and artificial intelligence algorithms responsible for processing images and making autonomous decisions in real time. High-intensity LEDs provide controlled and uniform illumination, reducing the effects of shadows and ambient light variations to improve detection reliability. An Arduino handles real-time sensor acquisition and motor control, receiving commands from the Raspberry Pi and ensuring fast and reliable communication between all subsystems.



LEDs



Locomotion system



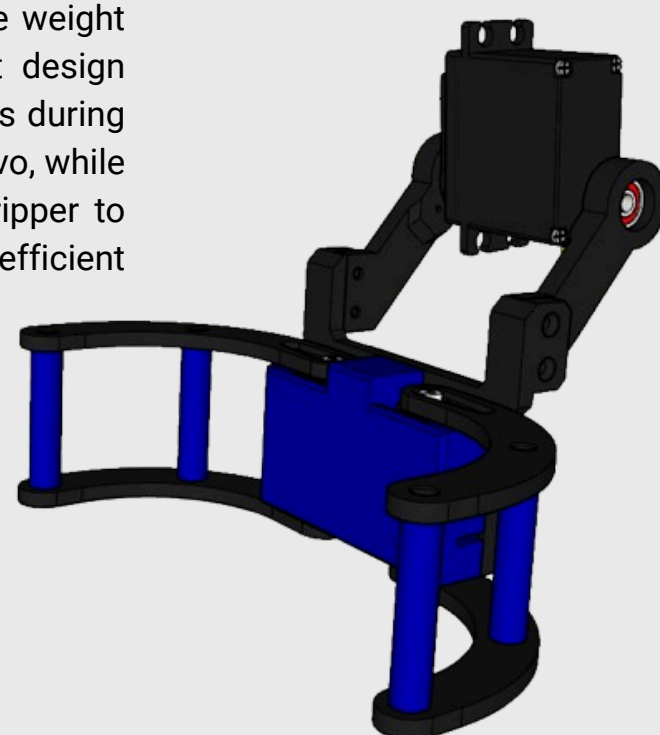
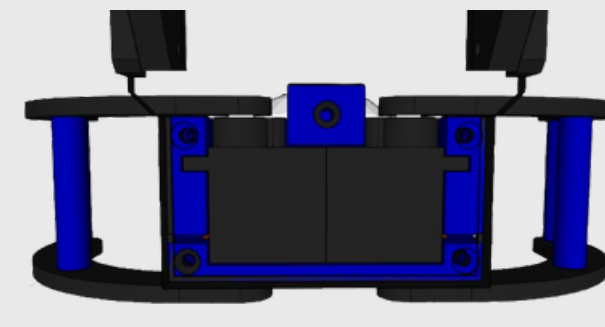
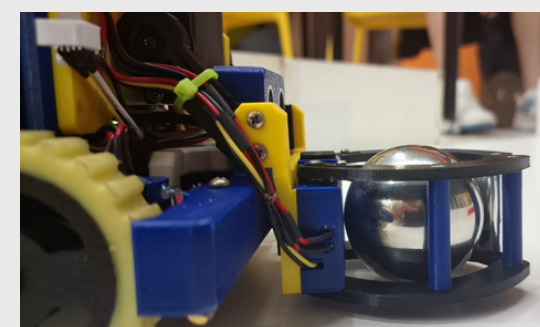
125° Zone camera



200° Line camera

COLLECTION MECHANISM

The rescue gripper was designed with a fully 3D-printed structure to minimize weight while maintaining the rigidity required for reliable operation. Its lightweight design reduces the load on the robot and enables faster and more precise movements during victim collection. The lifting mechanism is actuated by an S3003 (90–180°) servo, while two MG90S 9g metal gear servos control the opening and closing of the gripper to securely capture the rescue balls. This simple and compact design provides an efficient and reliable solution for the rescue task.



OUR HISTORY

We are Brainbots and we will be representing Brazil in Robocup Junior Rescue Line 2026! This is our fifth year in Robocup, and our second year participating internationally. With a total of three second places and one victory at our local district championship, one third place and one victory at Brazil championship, and participation at the last year's world championship, where we finished 7th, we are thrilled for this year's competition in Incheon!



Brasil Championship 2025



District Championship 2025



World Championship 2025



Brasil Championship 2024



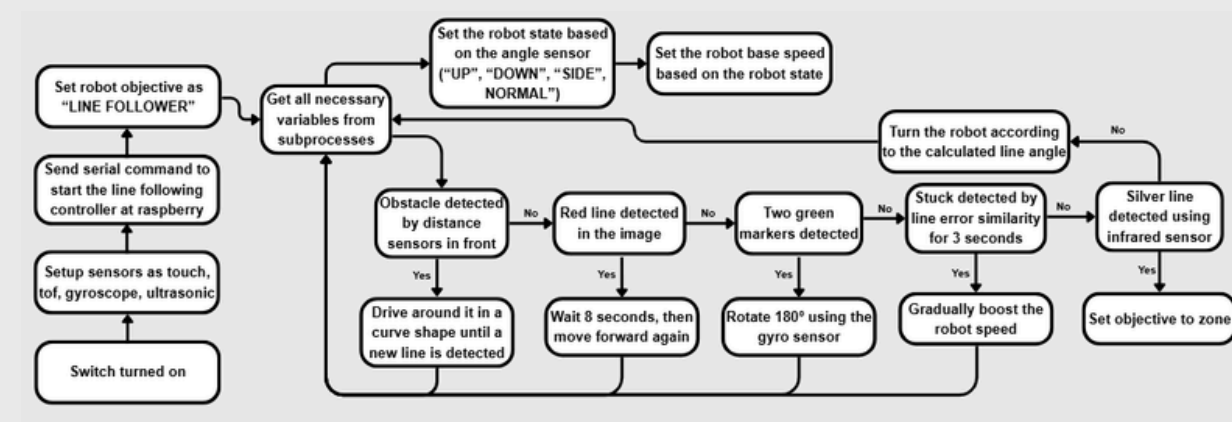
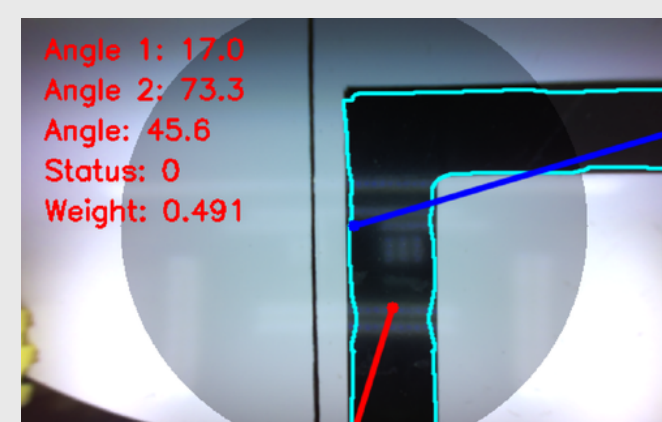
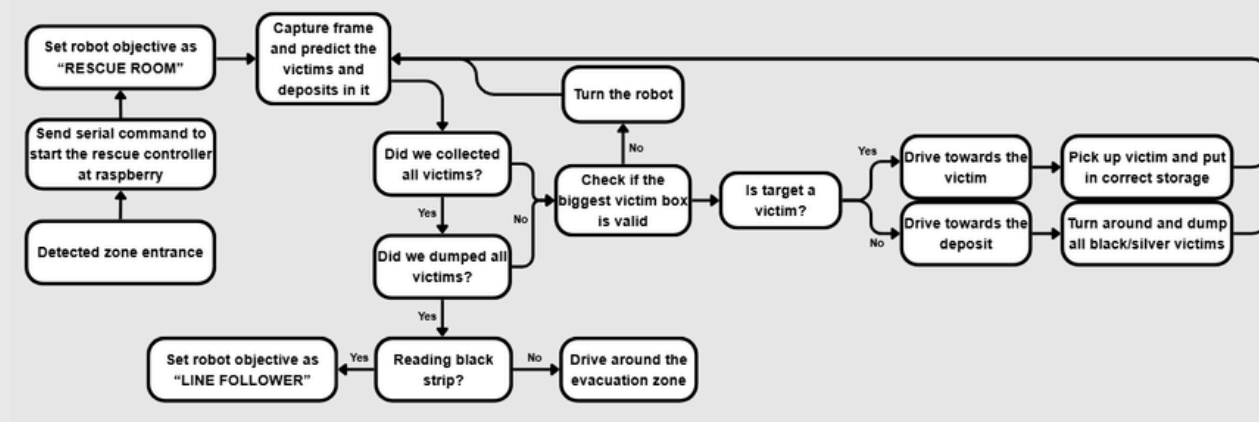
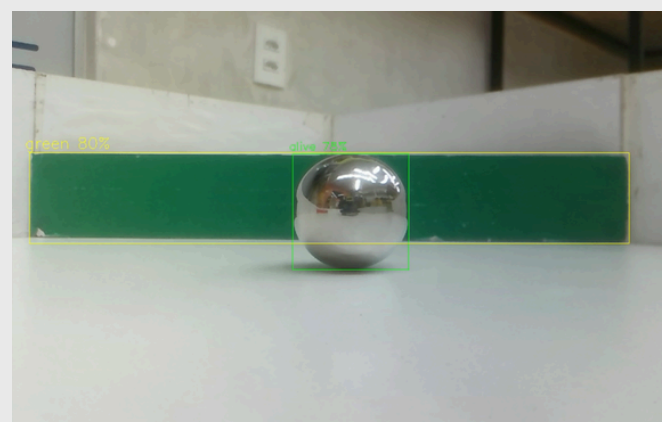
District Championship 2024



Brasil Championship 2023

EVACUATION ZONE

The evacuation-zone system is activated as soon as the dedicated silver sensor detects the entrance to the rescue area. At this point, the robot transitions from line-following mode to a vision-based rescue mode, using the victim-detection camera and an AI model to continuously identify victims and evacuation points. Once a victim is detected, the robot approaches it, performs the collection procedure, and stores it in its onboard compartment. Our strategy is to collect all victims before carrying out any delivery operation. Although this means the robot transports multiple victims simultaneously, it significantly reduces unnecessary movement inside the evacuation zone and decreases the overall rescue time. Victim detection is performed by a neural network accelerated by the Hailo-8 AI processor. Compared with traditional image-processing techniques, this AI-based approach proved more robust against variations in lighting, victim orientation, partial occlusions, and changes in camera perspective.



HARDWARE

3D MODELING

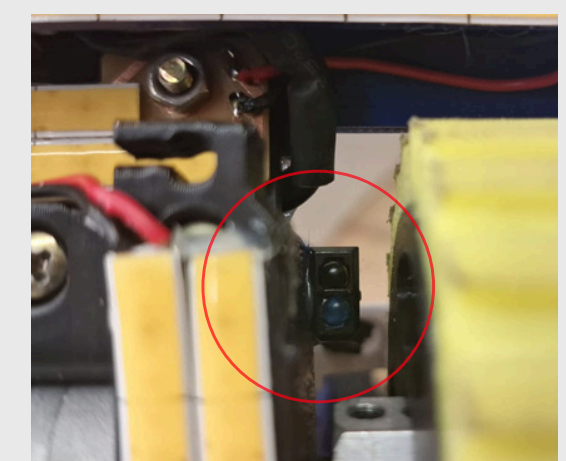


SketchUp

We chose SketchUp to design our robot because it provides an intuitive and efficient environment for 3D modeling, allowing rapid prototyping and easy modifications throughout the development process. Its user-friendly interface enabled the team to quickly test different layouts, verify component placement, and optimize the chassis before manufacturing. Additionally, the compatibility of the models with 3D printing workflows made SketchUp a practical and cost-effective solution that fit the needs of our project.

SILVER RECOGNITION

Recognizing the silver tile using the camera proved to be unreliable due to variations in lighting and surface reflections. To overcome this challenge, we adopted a dedicated hardware solution based on a silver detection sensor(TCRT 5000 IR Sensor), providing fast and consistent identification of the evacuation zone entrance. This approach increased reliability while allowing the vision system to focus on navigation and victim detection.



POWER SUPPLY

The robot is powered by six 18650 lithium batteries. Two batteries supply the Arduino, motors, sensors, and servos, while the remaining four are dedicated to the Raspberry Pi 5, providing a stable power source for the AI and computer vision systems and preventing interference between the processing and actuation subsystems.

